

Fixed Points and Stability

Lecture Notes

Topics: Fixed Points · Stability · Phase Line · Linearization

Where We Left Off

In the slope-field lesson we noticed that certain horizontal lines kept showing up. Last class we solved the equations and the same special values appeared again as long-time limits:

- Falling golf ball: $v(t) = \frac{g}{k}(1 - e^{-kt}) \rightarrow \frac{g}{k}$ as $t \rightarrow \infty$.
- Coffee cooling: $T(t) = T_a + (T_0 - T_a)e^{-kt} \rightarrow T_a$ as $t \rightarrow \infty$.

Opening question: what do those horizontal lines in the slope field tell us?

Answer: they sit at values of y where the slope is exactly 0. If the system starts there, it stays there. If it starts nearby, one of three things can happen:

- the system never reaches the value but gets steadily closer,
- the system falls away from the value,
- or (less common) it gets closer from one side and runs away on the other.

Today we give those special values a name (*fixed points*), and we develop two tests to decide which of the three behaviors happens.

Fixed Points

We work with autonomous first-order ODEs throughout: equations of the form

$$\frac{dy}{dt} = f(y).$$

The right-hand side depends only on the current value of y , not on the current time.

Fixed Point (Equilibrium)

A **fixed point** of the autonomous ODE $\frac{dy}{dt} = f(y)$ is a value y^* such that

$$f(y^*) = 0.$$

Equivalently: the constant function $y(t) \equiv y^*$ is a solution. If y ever equals y^* , then $y' = 0$, so y stays at y^* forever.

Finding fixed points: set the right-hand side to zero and solve for y . That's it.

Examples we've already seen:

- Golf ball $v' = g - kv$: setting $g - kv = 0$ gives $v^* = g/k$ (the terminal velocity).
- Cooling $T' = -k(T - T_a)$: setting $T - T_a = 0$ gives $T^* = T_a$ (room temperature).
- Exponential growth $y' = ky$: setting $ky = 0$ gives $y^* = 0$.

Stability

To understand a systems behavior, we need to know whether nearby solutions are pulled *toward* the fixed point or pushed *away*.

Stable and Unstable Fixed Points

A fixed point y^* of $y' = f(y)$ is

- **stable** if nearby solutions move *toward* y^* ,
- **unstable** if nearby solutions move *away from* y^* .

Note

Semi-stable fixed points are possible too: a fixed point can attract solutions from one side and repel them from the other. We won't focus on these in this course, but be aware they exist; they happen when f has a repeated root at y^* (the sign of f does not change as you cross y^*).

Two Tests for Stability

We will use two different tests for stability, and they always agree when both apply.

1. **Phase-line test:** look at the sign of $f(y)$ on either side of y^* . If f pushes y back toward y^* on both sides, the fixed point is stable. If f pushes y away on both sides, it is unstable.
2. **Linearization test:** look at the slope of f at y^* . If $f'(y^*) < 0$, stable. If $f'(y^*) > 0$, unstable. If $f'(y^*) = 0$, inconclusive; fall back on the phase line.

The phase-line test will always work in this class and does not require taking any derivatives. The linearization test shows us in more detail how solutions behave close to the fixed point and how quickly that behavior evolves.

The Phase Line

Phase Line

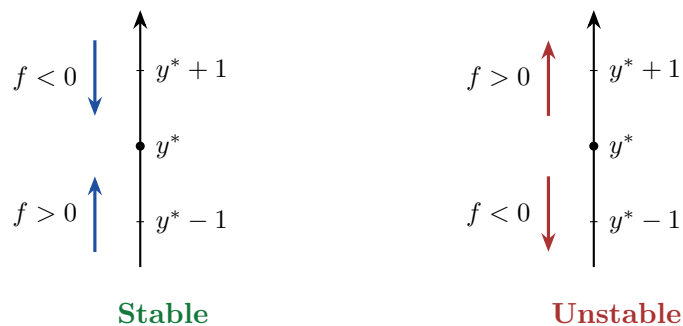
The **phase line** for an autonomous ODE $y' = f(y)$ is the y -axis with:

- fixed points marked as dots,
- up arrows in intervals where $f(y) > 0$ (solutions increase),
- down arrows in intervals where $f(y) < 0$ (solutions decrease).

It's the slope field with the time axis collapsed. Because the equation is autonomous, every horizontal slice of the slope field looks the same, so we lose nothing by collapsing.

What Stable and Unstable Look Like

The two pictures below show how arrows around a single fixed point y^* determine its stability.



Note

The phase-line test never fails: it only uses the sign of f , which we can always read off. When the linearization test gives $f'(y^*) = 0$ (inconclusive), or when we just want a quick check, the phase line is what we fall back on.

Why Linearization Works

The linearization test is so quick (just check the sign of $f'(y^*)$) that it can feel like a black-box rule. Below we show that it comes straight from Taylor expanding f near y^* , and we will see that it tells us about what happens to the gap between a solution and the fixed point over time.

The question we want to answer in this section is: when we start just a little bit away from a fixed point y^* , how does the gap between $y(t)$ and y^* change over time?

Set Up: The Displacement from the Fixed Point

Suppose $y(t)$ starts a little away from y^* . Let

$$u(t) = y(t) - y^*$$

be the **displacement** from the fixed point. Because y^* is a constant,

$$\frac{du}{dt} = \frac{dy}{dt} = f(y) = f(y^* + u).$$

So u satisfies its own ODE: $u' = f(y^* + u)$.

Apply Taylor Around y^*

Taylor expand f around $a = y^*$, evaluated at $x = y^* + u$ (so $x - a = u$):

$$f(y^* + u) = f(y^*) + f'(y^*)u + \frac{f''(y^*)}{2!}u^2 + \frac{f'''(y^*)}{3!}u^3 + \dots$$

Since y^* is a fixed point, $f(y^*) = 0$. For u small, u^2 , u^3 , ... shrink *much* faster than u . So for solutions close to y^* ,

$$f(y^* + u) \approx f'(y^*)u.$$

Result: remember Friday?

Substituting back:

$$\frac{du}{dt} \approx f'(y^*) u.$$

Set $k = f'(y^*) \implies u' = ku \implies$

$$u(t) \approx u(0) e^{f'(y^*) t}$$

This is the exponential growth/decay equation from last class.

Read Off Stability from the Sign

The sign of $f'(y^*)$ controls everything:

- $f'(y^*) < 0$: the exponent is negative, $u(t) \rightarrow 0$. The gap shrinks. **Stable.**
- $f'(y^*) > 0$: the exponent is positive, $|u(t)| \rightarrow \infty$. The gap grows exponentially. **Unstable.**
- $f'(y^*) = 0$: we've thrown away all the information we need; the linear approximation is just a tangent line that touches zero without crossing. **Inconclusive.**

Note

Linearization tells you more than stability: it also tells you the *rate*: if $f'(y^*) = -2$, the gap to the fixed point shrinks like e^{-2t} ; if $f'(y^*) = -10$, it shrinks like e^{-10t} , ten times faster. The phase-line test gives you stability (yes/no); linearization gives you stability and a time constant.

Example 1: The Falling Golf Ball

$$\frac{dv}{dt} = g - kv, \text{ with } g, k > 0$$

Step 1: Find the fixed points: set $g - kv = 0$:

$$v^* = \frac{g}{k} = v_{\text{term}}.$$

One fixed point, the terminal velocity.

Step 2: Phase line: with $f(v) = g - kv$:

- $v < v_{\text{term}}$: $g - kv > 0$. Up arrow. (Speed grows.)
- $v > v_{\text{term}}$: $g - kv < 0$. Down arrow. (Speed shrinks.)

Both arrows point toward v_{term} : **stable.**

Step 3: Linearization check: $f'(v) = -k$, so $f'(v^*) = -k < 0$. **Stable.** ✓

The linearization rate is k : the gap to terminal velocity shrinks like e^{-kt} . That matches the closed-form solution

$$v(t) = v_{\text{term}}(1 - e^{-kt}) \iff v_{\text{term}} - v(t) = v_{\text{term}} e^{-kt}$$

which we derived last class. The displacement $u(t) = v(t) - v_{\text{term}}$ decays exactly as the linearization predicts.

Note

Three views of the same answer: the slope field showed a horizontal line at v_{term} with arrows funneling onto it. The closed-form solution showed an exponential approach to v_{term} . The phase line and linearization show the same approach in one line each. The phase line is the cheapest of the three: no integration, no figure, just zeros of f and signs around them.

Example 2: A Quadratic Phase Line

A common pattern shows up whenever f is a polynomial: the fixed points are roots of f , and stability often *alternates* along the y -axis.

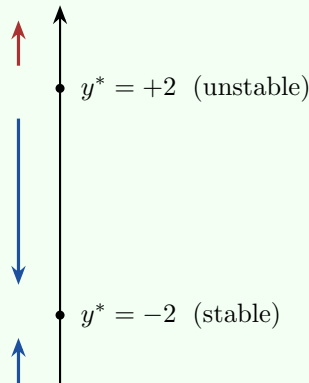
$$\frac{dy}{dt} = y^2 - 4$$

Step 1: Find the fixed points: factor:

$$y^2 - 4 = (y - 2)(y + 2) = 0 \implies y^* = +2 \text{ or } y^* = -2.$$

Step 2: Phase line: check the sign of $f(y) = (y - 2)(y + 2)$ in each region:

Region	Sign of $(y - 2)(y + 2)$	Arrow
$y < -2$	$(-)(-) = +$	up
$-2 < y < 2$	$(-)(+) = -$	down
$y > 2$	$(+)(+) = +$	up



Step 3: Classify:

- $y^* = -2$: arrows on both sides point in. **Stable.**
- $y^* = +2$: arrows on both sides point out. **Unstable.**

Step 4: Linearization check: $f'(y) = 2y$.

- $f'(-2) = -4 < 0$: stable. ✓
- $f'(2) = +4 > 0$: unstable. ✓

What the rate tells us: near $y^* = +2$, displacements grow like $u(t) = u(0)e^{4t}$, so the gap multiplies by $e^4 \approx 54.6$ per unit time. That's fast. Near $y^* = -2$, displacements shrink like $u(t) = u(0)e^{-4t}$, so the gap is cut by e^4 per unit time.

Note

Alternating stability is the generic pattern: when f is a polynomial with simple roots and a positive leading coefficient, sign changes from one root to the next, so stability alternates along the y -axis. The same logic explains why the logistic equation $P' = rP(1 - P/K)$ has an unstable fixed point at 0 and a stable one at K .

Summary

- **Fixed point:** y^* with $f(y^*) = 0$. Equivalently, a constant solution of $y' = f(y)$.
- **Stability:** a fixed point is stable if nearby solutions move toward it, unstable if they move away.
- **Phase-line test:** sign of $f(y)$ on either side of y^* . Always works; tells stability.
- **Linearization test:** sign of $f'(y^*)$. Negative $\Rightarrow$ stable, positive $\Rightarrow$ unstable, zero $\Rightarrow$ inconclusive. Bonus: $|f'(y^*)|$ is the rate of approach/escape.
- **Why linearization works:** Taylor expand f around y^* . With $u = y - y^*$,

$$u' = f(y^* + u) \approx f'(y^*)u \implies u(t) \approx u(0)e^{f'(y^*)t}.$$

Every fixed point is locally exponential; the sign of $f'(y^*)$ decides the direction.